

Customer Insights & Loan Process Optimization

1. Problem Understanding

A multinational bank is experiencing declining personal banking adoption. The root cause is threefold: manual, document-heavy loan and onboarding workflows that create delays; siloed customer data that prevents proactive retention analytics; and an absence of ML infrastructure that could enable personalized, competitive offerings. Underwriters currently retrieve and review documents by hand, making it impossible to scale or act on insights in near real-time.

The core technical challenge is bridging unstructured, multi-format documents (bank statements, onboarding forms, loan applications across regions and templates) with a governed, analytics-ready data platform. Without this bridge, neither ML-driven decisioning nor executive dashboards are achievable.

2. Clarifying Questions for the Client

Area	Question
Data & Volumes	What is the estimated monthly volume of documents ingested? Are they already digitised (PDF/image scans) or still paper-based?
Regulatory	Which jurisdictions does the bank operate in? Are there data-residency constraints (e.g. GDPR, RBI, MAS) that restrict cross-border data movement?
Integration	What are the existing core banking systems (e.g. Temenos, Finacle)? Is there an existing API gateway or ESB?
ML Maturity	Does the bank have a data science team? Is there a preference for a managed ML platform (e.g. SageMaker, Vertex AI) or on-premise MLflow?
SLAs	What is the acceptable latency for underwriter document retrieval? Is a sub-second query response expected?
Identity	Is there a single customer identifier across banking products, or do silos maintain separate IDs?

3. Proposed Solution Summary

I propose a cloud-native, multi-agent Intelligent Document Processing (IDP) and Analytics platform built on AWS (with equivalent on-prem alternatives noted). The solution has three logical layers:

- Intelligent Ingestion Layer: autonomous agents extract and structure information from multi-template documents using LLM-backed OCR and Named Entity Recognition (NER).
- Unified Data Platform: a governed lakehouse (raw, curated, and feature-store zones) feeds both real-time operational queries and batch ML training pipelines.
- Insight & Action Layer: low-latency dashboards for underwriters and executives, plus ML models for churn prediction, credit risk scoring, and personalised product recommendations.

Key benefits: 60-70% reduction in manual document handling time, a single customer 360 view across products, and a retrainable ML backbone that improves over time. The architecture is OWASP ASVS Level 2 compliant by design, with PII tokenisation, field-level encryption, and role-based access at every layer.

4. Detailed Solution Architecture

4a. Architecture Diagram

See Mermaid diagram in the companion file. Key components below.

Layer	Components & AWS Services
Document Ingestion	S3 (raw landing), Textract (OCR/forms), Comprehend (NER/PII detection), Lambda (orchestration), SQS (async queuing). On-prem: Apache NiFi + Tesseract + spaCy.
Multi-Agent MCP Layer	Agent Controller (orchestrates sub-agents), Document Classifier Agent, Extraction Agent, Validation Agent, and Routing Agent. Agents communicate via MCP (Model Context Protocol) over a secured internal bus. LLM backbone: Claude claude-sonnet-4-6 via Bedrock.
Data Platform	S3-based lakehouse (Bronze/Silver/Gold zones), AWS Glue (ETL), Apache Iceberg (table format for schema evolution), Redshift Serverless (BI queries), RDS Aurora (operational store for underwriters).
Feature Store & ML	SageMaker Feature Store, SageMaker Pipelines (training/retraining), Model Registry, A/B endpoint deployment. Models: churn prediction (XGBoost), credit risk (LightGBM), product recommendation (two-tower neural net).
Insight Layer	Amazon QuickSight (executive dashboards), custom React portal for underwriters querying Aurora, Elasticsearch for full-text document retrieval (<200ms).
Security (OWASP ASVS L2)	Macie (PII scanning), KMS (field-level encryption), IAM + Cognito (AuthN/AuthZ), API Gateway with WAF, VPC with private subnets, CloudTrail + GuardDuty (audit & threat detection). All secrets in Secrets Manager.

4b. Architectural Justification

- Lakehouse over traditional DWH: Iceberg table format allows schema evolution as new document templates are introduced, without breaking downstream ML pipelines.
- Multi-agent MCP architecture: separating classification, extraction, validation, and routing into specialised agents improves accuracy and makes the system auditable. Each agent can be independently retrained or swapped.
- Async ingestion via SQS: decouples document arrival spikes from processing, ensuring no data loss during peak submission periods.
- OWASP ASVS Level 2 chosen over Level 3 for initial phases: Level 3 requires formal code review and penetration testing gates that add 4-6 weeks per release; Level 2 provides strong controls (MFA, encrypted transit/rest, session management, input validation) appropriate for a regulated bank without blocking velocity.

5. Phase-Wise Roadmap

The delivery is structured across three phases to manage risk and demonstrate value early. The Proof of Concept runs for six weeks with a lean team of four, targeting a single document type to validate extraction accuracy above 85% and prove the end-to-end pipeline. The Pilot phase spans three months, expanding to all document types, all multi-agent MCP components, a first churn prediction model, and the underwriter portal, with OWASP ASVS Level 2 controls in place and a VAPT sign-off required before go-live. Full Production rollout follows over the next three months with a team of twelve, covering all geographies, the complete ML suite including credit risk scoring and product recommendations, auto-retraining pipelines, and disaster recovery, targeting 99.9% uptime and a 50% reduction in loan processing time as the primary exit criterion.